KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 27 · MATH

The Planetarium Number Hunt

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Counting Zeros

Kiona is helping at the Adler Planetarium in Chicago, Illinois, where space numbers have lots of zeros. Warm up before the lesson, and show your work. Some problems may feel new, and that is okay!

1. Review (Week 26): evaluate each expression. a) 5 + 8 × 4; b) 3 × [16 − (2 + 9)]

2. Find each product. a) 10 × 10; b) 10 × 10 × 10; c) 10 × 10 × 10 × 10. What pattern do you see in the zeros?

3. Multiply: 26 × 10 and 26 × 100. Explain what happens to the digits 2 and 6.

4. Which number is 10 times as large as 4.7?

   47 0.47 4.70

5. Kimani says 10 × 10 × 10 × 10 × 10 is 50. Do you agree? Explain your thinking.

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The Planetarium Number Hunt

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Challenge: Space Number Check

The exponent counts the tens. Multiplying moves digits to the left, and dividing moves digits to the right. Show your work.

1. Write each in standard form. a) 10^5; b) 7×10^3; c) $4 \times 10^4 + 6 \times 10^2$

2. Multiply or divide. a) 2.36×10^2; b) 0.08×10^3; c) $5{,}400 \div 10^3$; d) $61 \div 10^2$

3. Two steps: a gift shop orders 10^3 moon keychains at $1.25 each and 10^2 planet magnets at $0.90 each. What is the total cost?

4. Find the mistake: Kirra says $3.9 \times 10^2 = 3.900$. In two or three complete sentences, explain her mistake and give the correct product.

★ **Challenge:** 10^7 is how many times as large as 10^3? Then write $1{,}000 \times 10{,}000$ as one power of 10, and explain how you know.

Looking Ahead: Week 28: Multiplying and Dividing Decimals. At a cross-country running meet in Boulder, Colorado, you will multiply and divide decimals to hundredths with times, distances, and pace.

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 27 · READING

The Quiet Clue

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Think Like a Detective

Today you will solve two mysteries set in Savannah, Georgia: one in a bookshop on Wright Square and one in Forsyth Park. Show what you already know. It is fine to guess!

1. Review (Week 26): Mitsuko says, "Fractions are my Achilles' heel." What does she mean? Tell the myth behind the allusion in one sentence.

2. Name a mystery book, show, or movie you know. What was the puzzle, and who solved it?

3. Read: "The window was open, and the goldfish bowl was empty. The cat sat on the sill. But the fish was swimming in the kitchen sink, where Mom was cleaning the bowl." Who looked guilty? What was the truth?

4. A **red herring** in a mystery is most likely a clue that...

   points to the truth | points the wrong way | is a real fish

5. Why do you think a detective should look at ALL the clues before naming a suspect?

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 27 · READING

The Quiet Clue

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Case Notes

Write in complete sentences with correct capitals, commas, and quotation marks.

1. Read: "Ms. Moira's reading glasses were missing. Mikael had ink on his fingers. A pencil was tucked behind Ms. Moira's ear, and a shiny frame peeked out of her hair." Name the red herring, the real clue, and your conclusion.

2. Two steps. In "The Case of the Missing Compass," which clue does Mitsuko call quiet? Explain how it narrows the suspects to two people.

3. Find the mistake. Monty wrote: "In both mysteries, the first suspect is guilty." Correct his sentence and give one piece of evidence from each story.

4. Be the editor. Rewrite each sentence correctly. a) "Moxie is innocent" Mrs. Mitra said. b) The pies was in the cooler.

★ **Challenge:** Write the opening of your own short mystery set in Savannah, Georgia, with one red herring and one quiet clue (3 or 4 sentences).

Looking Ahead: Week 28: Reading several sources to answer one question. You will gather facts about the monarch butterfly migration to the forests of Michoacán, Mexico, take notes, and cite which source said what.

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 27 · SCIENCE

Earth's Water by the Numbers

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. **Review (Week 26):** Explain the difference between mass and weight. Then tell what a 10-kilogram bag weighs on Earth (9.8 newtons on each kilogram).

2. Predict: Out of every 100 cups of Earth's water, how many cups do you think are fresh water? Explain your guess.

3. Where do you think most of Earth's fresh water is: lakes, rivers, ice or underground? Why?

4. Name three ways your family uses water in one day. Which one do you think uses the most?

5. Draw a quick graph of anything you like to count (pets, books or rainy days). What kind of graph did you choose?

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 27 · SCIENCE

Earth's Water by the Numbers

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Counting Every Drop

Use evidence from today's lesson. Write in complete sentences.

1. **Compare.** Fresh groundwater is about 2,526,000 cubic miles; fresh lakes are about 21,830. Which is bigger, and about how many times bigger?

2. **Honest bar graph.** Graph 2015 U.S. water use: thermoelectric 133, irrigation 118, public supply 39 (billion gallons per day). Start at zero, label it and add a title.

3. **Two-step.** Turning off the tap while brushing saves about 8 gallons a day. How much for a family of 4 in one week?

4. **Find the mistake.** "Rivers hold most of Earth's fresh water." Explain what is wrong using the USGS numbers.

★ **Challenge: Explain.** How can the GRACE satellites tell that groundwater is shrinking, using gravity?

Looking Ahead: Next week: Heat and Thermal Energy. You will investigate how heat moves by conduction, convection and radiation, and test insulators.

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 27 · SOCIAL STUDIES

More Voices for Equal Rights (1960s–1970s)

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Write what you know right now. Guesses are fine; this is your starting point.

1. **Review (Week 26):** Name one tool the civil rights movement used (a court case, a boycott, a march or voter registration) and one law it helped win.

2. Besides Black Americans, which other groups do you think worked for equal rights in the 1960s and 1970s? Write two guesses.

3. Prediction: What is the difference between a strike and a boycott? Guess how workers could use both at once.

4. Do you think girls and boys always had the same chances to play school sports? Explain your thinking.

5. Write one question you have about the fight for equal rights.

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More Voices for Equal Rights (1960s–1970s)

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Many Voices, One Pattern

Use evidence from today's lesson. Write in complete sentences.

1. **Put in order with years:** Section 504 sit-in; Title IX; Alcatraz occupation begins; Delano grape strike begins; Section 504 passes.

2. **Find the mistake.** "Title IX is a law only about school sports." Rewrite the sentence so it is true.

3. **Compare two sources.** Title VI (1964) protects people "on the ground of race, color, or national origin." Section 504 (1973) uses almost the same words for disability. What does that show?

4. **Write a paragraph.** Choose Dolores Huerta, Larry Itliong, Billy Frank Jr., Patsy Takemoto Mink or Judy Heumann. Write 3 or 4 sentences about what this person was shut out of and how they responded. Check capitals, commas and end punctuation.

★ **Challenge: Two steps:** The Alcatraz occupation began in November 1969 and lasted 19 months. In what month and year did it end? Show your work.

Looking Ahead: Next week: the Cold War and the Space Race (1945–1991), from Sputnik and Apollo to Katherine Johnson and the fall of the Berlin Wall.